

3R (Articulated)

Key Characteristics:

The manipulator under study is a three-revolute-joint (3R) serial robotic arm. All three joints are actuated revolute joints, making this a fully rotational, open-chain mechanism with three degrees of freedom. The system is designed for educational demonstration of robotic kinematics, dynamics, and point-to-point positioning.

Table 1: Key Characteristics of the 3R Manipulator

Parameter	Value / Description
Configuration	RRR — three revolute joints connected in series (serial chain)
Degrees of Freedom (DOF)	3
Joint 1 — θ_1	Rotation about vertical Z_0 axis (base rotation), range: $0^\circ - 180^\circ$
Joint 2 — θ_2	Rotation about horizontal Z_1 axis (shoulder), range: $0^\circ - 180^\circ$
Joint 3 — θ_3	Rotation about Z_2 axis (elbow), range: $0^\circ - 180^\circ$
Base Height (d_1)	90 mm — vertical offset from ground to shoulder joint centre
Link 2 Length (l_2)	210 mm
Link 3 Length (l_3)	170 mm (includes end-effector structural body)
Maximum Vertical Reach	$d_1 + l_2 + l_3 = 90 + 210 + 170 = 470$ mm (arm fully extended upward, $\theta_2=90^\circ$, $\theta_3=0^\circ$)
Minimum Vertical Reach	$d_1 + l_2 - l_3 = 90 + 210 - 170 = 130$ mm (arm folded back, $\theta_2=90^\circ$, $\theta_3=180^\circ$)
End-Effector	Point at tip of Link 3 — no gripper; gripper profile is structural only
Actuators	MG995 Metal Gear Servo Motors $\times 3$ — torque 9.4 kg·cm @ 6 V
Controller	ESP32 Microcontroller — PWM via ledc peripheral (14-bit resolution)
Link Material	Acrylic (PMMA) sheet — 5 mm thick, $\rho \approx 1180$ kg/m ³
Link Cross-Section	Width (z-axis) = 60 mm, Height (y-axis) = 55 mm for Links 2 & 3
Link 1 Cross-Section	Width (z-axis) = 60 mm, Horizontal (x-axis) = 55 mm, Height (y-axis) = 90 mm
Design Software	SolidWorks (3D CAD and assembly)
Analysis Software	MATLAB (kinematics simulation, workspace, plots)
Firmware IDE	Arduino IDE (ESP32 PWM servo control)

DH Frame Assignment

Coordinate frames are assigned following the standard Denavit-Hartenberg convention. The Z-axis of each frame coincides with the joint rotation axis. The symbol $\otimes$ in the design sketch denotes a Z-axis directed into the page. Z_0 points vertically upward. Z_1 and Z_2 are horizontal and parallel (both pointing out of the plane in the side-view sketch), consistent with $\alpha_1 = 90^\circ$ at joint 1 and $\alpha_2 = \alpha_3 = 0^\circ$.

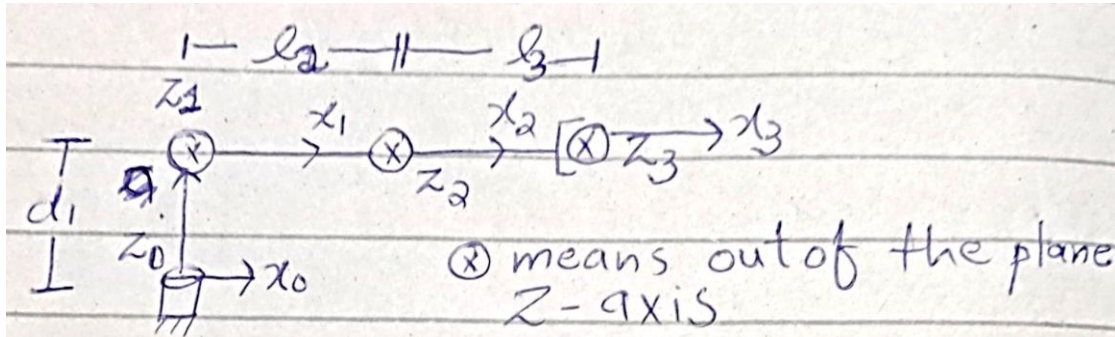


Figure 2: DH Frame Assignment Sketch — Side View showing Z_0, Z_1, Z_2 axes and O_0, O_1, O_2, O_3 origins

Home / Reference Configuration

Before executing any motion task, the manipulator is driven to a fixed reference pose known as the home configuration. This provides a consistent, safe starting state. The home joint angles are:

$$\theta_1 = 0^\circ, \theta_2 = 0^\circ, \theta_3 = 90^\circ$$

Substituting into the forward kinematics (derived in Section 4):

$$x = \cos 0^\circ \cdot (210 \cos 0^\circ + 170 \cos 90^\circ) = 1 \cdot (210 + 0) = 210 \text{ mm}$$

$$y = \sin 0^\circ \cdot (210 \cos 0^\circ + 170 \cos 90^\circ) = 0 \text{ mm}$$

$$z = 90 + 210 \sin 0^\circ + 170 \sin 90^\circ = 90 + 0 + 170 = 260 \text{ mm}$$

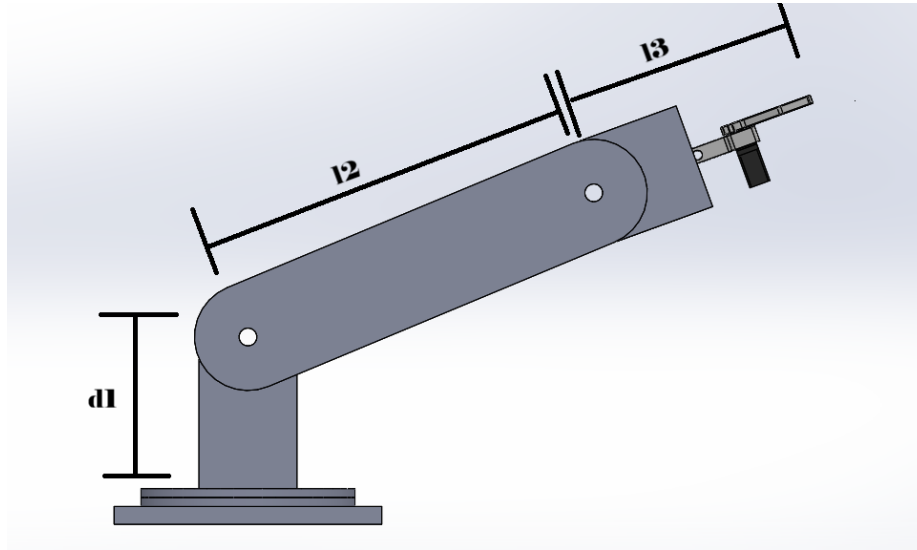
Home end-effector position: $O_3 = (210, 0, 260)$ mm in the base frame.

Table 2: PWM Initialisation Values at Home Configuration

Joint	Home Angle	PWM Pulse Width
Joint 1 (θ_1)	0°	500 μ s
Joint 2 (θ_2)	0°	500 μ s

Joint 3 (θ_3)	90°	1500 μ s
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PWM frame frequency: 50 Hz. Range: 500 μ s (0°) to 2500 μ s (180°). On power-up, the ESP32 drives all channels to home values before accepting motion commands.



Homogeneous Transformation Matrices

A homogeneous transformation matrix $T \in \mathbb{R}^{4 \times 4}$ encodes both the rotation and translation between two consecutive coordinate frames. Using the standard Denavit-Hartenberg convention, the transformation from frame $i-1$ to frame i is:

$$T_m = Rot(Z, \theta_i) \cdot Trans(Z, d_i) \cdot Trans(X, a_i) \cdot Rot(X, \alpha_i)$$

Expanding into 4×4 matrix form, where $c\theta = \cos\theta$, $s\theta = \sin\theta$, $c\alpha = \cos\alpha$, $s\alpha = \sin\alpha$:

$$T_i = \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The shorthand notation $c_i = \cos\theta_i$, $s_i = \sin\theta_i$, $c_{23} = \cos(\theta_2 + \theta_3)$, $s_{23} = \sin(\theta_2 + \theta_3)$ is used throughout.

Individual Matrix ${}^0T^1$ ($a_1=0$, $\alpha_1=90^\circ$, $d_1=90$, θ_1)

Substituting $a_1=0$, $\alpha_1=90^\circ \Rightarrow \cos\alpha_1=0$, $\sin\alpha_1=1$, $d_1=90$ mm:

$${}^0T^1 = \begin{bmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & 90 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Individual Matrix ${}^1T^2$ ($a_2=210$, $\alpha_2=0^\circ$, $d_2=0$, θ_2)

Substituting $a_2=210$ mm, $\alpha_2=0^\circ \Rightarrow \cos\alpha_2=1$, $\sin\alpha_2=0$, $d_2=0$:

$${}^1T^2 = \begin{bmatrix} c_2 & -s_2 & 0 & 210c_2 \\ s_2 & c_2 & 0 & 210s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Individual Matrix ${}^2T^3$ ($a_3=170$, $\alpha_3=0^\circ$, $d_3=0$, θ_3)

Substituting $a_3=170$ mm, $\alpha_3=0^\circ$, $d_3=0$:

$${}^2T^3 = \begin{bmatrix} c_3 & -s_3 & 0 & 170c_3 \\ s_3 & c_3 & 0 & 170s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Cumulative Matrix ${}^0T^2 = {}^0T^1 \cdot {}^1T^2$

Performing full row-by-column multiplication. Key element computations:

$$R(1,1) = c_1 \cdot c_2 + 0 \cdot s_2 + s_1 \cdot 0 = c_1c_2$$

$$R(2,1) = s_1 \cdot c_2 + 0 \cdot s_2 + (-c_1) \cdot 0 = s_1c_2$$

$$R(3,1) = 0 \cdot c_2 + 1 \cdot s_2 + 0 \cdot 0 = s_2$$

$$R(1,2) = c_1 \cdot (-s_2) + 0 \cdot c_2 + s_1 \cdot 0 = -c_1s_2$$

$$R(1,3) = c_1 \cdot 0 + 0 \cdot 0 + s_1 \cdot 1 = s_1$$

$$R(2,3) = s_1 \cdot 0 + 0 \cdot 0 + (-c_1) \cdot 1 = -c_1$$

$$p_x = c_1 \cdot 210c_2 + 0 \cdot 210s_2 + s_1 \cdot 0 + 0 = 210c_1c_2$$

$$p_y = s_1 \cdot 210c_2 + 0 \cdot 210s_2 + (-c_1) \cdot 0 + 0 = 210s_1c_2$$

$$p_z = 0 \cdot 210c_2 + 1 \cdot 210s_2 + 0 \cdot 0 + 90 = 210s_2 + 90$$

Full result:

$${}^0T^2 = \begin{bmatrix} c_1c_2 & -c_1s_2 & s_1 & 210c_1c_2 \\ s_1c_2 & -s_1s_2 & -c_1 & 210s_1c_2 \\ s_2 & c_2 & 0 & 210s_2 + 90 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Full End-Effector Transform ${}^0T^3 = {}^0T^2 \cdot {}^2T^3$

Using the angle-sum identities: $c_2c_3 - s_2s_3 = c_{23}$ and $s_2c_3 + c_2s_3 = s_{23}$. Key element derivations:

$$R(1,1) = c_1c_2 \cdot c_3 + (-c_1s_2) \cdot s_3 + s_1 \cdot 0 = c_1(c_2c_3 - s_2s_3) = c_1c_{23}$$

$$R(1,2) = c_1c_2 \cdot (-s_3) + (-c_1s_2) \cdot c_3 = -c_1(c_2s_3 + s_2c_3) = -c_1s_{23}$$

$$R(2,1) = s_1c_2c_3 - s_1s_2s_3 = s_1c_{23}$$

$$R(3,1) = s_2c_3 + c_2s_3 = s_{23}$$

$$p_x = 170c_1c_{23} + 210c_1c_2 = c_1(210c_2 + 170c_{23})$$

$$p_y = 170s_1c_{23} + 210s_1c_2 = s_1(210c_2 + 170c_{23})$$

$$p_z = 170(s_2c_3 + c_2s_3) + 210s_2 + 90 = 90 + 210s_2 + 170s_{23}$$

Final result:

$${}^0T^3 = \begin{bmatrix} c_1c_{23} & -c_1s_{23} & s_1 & c_1(210c_2 + 170c_{23}) \\ s_1c_{23} & -s_1s_{23} & -c_1 & s_1(210c_2 + 170c_{23}) \\ s_{23} & c_{23} & 0 & 90 + 210s_2 + 170s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Forward Position Kinematics:

Forward kinematics maps a set of joint angles ($\theta_1, \theta_2, \theta_3$) to the Cartesian position of the end-effector point O_3 in the base frame. The position is extracted from the fourth column of ${}^0T^3$.

DH Parameter Table

Table 3: Denavit-Hartenberg Parameters

Joint i	a_i (mm)	α_i (deg)	d_i (mm)	θ_i
1	0	90°	$d_1 = 90$	θ_1^* ($0^\circ - 180^\circ$)
2	$l_2 = 210$	0°	0	θ_2^* ($0^\circ - 180^\circ$)
3	$l_3 = 170$	0°	0	θ_3^* ($180^\circ - 180^\circ$)

The asterisk (*) marks the variable (active) parameter. All other entries per row are constant geometric properties of the link.

Forward Kinematics Equations

Extracting the position column of ${}^0T^3$, the end-effector position $O_3 = (x, y, z)$ in the base frame is:

$$\begin{aligned}x &= c_1 \cdot (210c_2 + 170c_{23}) \\y &= s_1 \cdot (210c_2 + 170c_{23}) \\z &= 90 + 210s_2 + 170s_{23}\end{aligned}$$

Physical interpretation: x and y share the same horizontal planar radius $r = 210c_2 + 170c_{23}$, scaled by $\cos\theta_1$ and $\sin\theta_1$ respectively. Joint 1 sweeps the arm through the horizontal plane; joints 2 and 3 control elevation.

Velocity Kinematics:

Velocity kinematics relates joint angular velocities to the end-effector linear and angular velocities through the Jacobian matrix. The instructor's explicit geometric Jacobian method is followed throughout, using skew-symmetric matrix notation $S(Z)$ in place of direct cross products.

Jacobian Method Overview

For each revolute joint i , the Jacobian column is:

$$\begin{aligned}J_{vi} &= S(Z_{i-1}) \cdot (O_3 - O_{i-1}) \quad [\text{linear velocity contribution}] \\J_{\omega i} &= Z_{i-1} \quad [\text{angular velocity contribution}]\end{aligned}$$

where $S(Z)$ is the skew-symmetric matrix of $Z = [z_x, z_y, z_z]^T$:

$$S(Z) = \begin{bmatrix} 0 & -z_z & z_y \\ z_z & 0 & -z_x \\ -z_y & z_x & 0 \end{bmatrix}$$

The end-effector position is denoted O_3 (not P). All subtractions $O_3 - O_i$ are performed term-by-term without premature cancellation.

Step 1: Z-Axis Vectors (from 0T_i rotation column 3)

$$Z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$Z_1 = \text{Third column of } R \text{ in } {}^0T^1 = \begin{bmatrix} s_1 \\ -c_1 \\ 0 \end{bmatrix}$$

$$Z_2 = \text{Third column of } R \text{ in } {}^0T^2 = \begin{bmatrix} s_1 \\ -c_1 \\ 0 \end{bmatrix}$$

Note: $Z_1 = Z_2$ because $\alpha_2 = 0^\circ$ introduces no twist between frames 1 and 2.

Step 2: Origin Vectors (from 0T_i position column 4)

$$O_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$O_1 = \begin{bmatrix} 0 \\ 0 \\ 90 \end{bmatrix}$$

$$O_2 = \begin{bmatrix} 210c_1c_2 \\ 210s_1c_2 \\ 90 + 210s_2 \end{bmatrix}$$

$$O_3 = \begin{bmatrix} c_1(210c_2 + 170c_{23}) \\ s_1(210c_2 + 170c_{23}) \\ 90 + 210s_2 + 170s_{23} \end{bmatrix}$$

Step 3: Skew-Symmetric Matrices $S(Z)$

$S(Z_0)$: $Z_0 = [0, 0, 1]^T \Rightarrow z_x=0, z_y=0, z_z=1$

$$S(Z_0) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$S(Z_1)$: $Z_1 = [s_1, -c_1, 0]^T \Rightarrow z_x=s_1, z_y=-c_1, z_z=0$

$$S(Z_1) = \begin{bmatrix} 0 & 0 & -c_1 \\ 0 & 0 & -s_1 \\ c_1 & s_1 & 0 \end{bmatrix}$$

$S(Z_2) = S(Z_1)$ (same vector)

$$S(Z_2) = \begin{bmatrix} 0 & 0 & -c_1 \\ 0 & 0 & -s_1 \\ c_1 & s_1 & 0 \end{bmatrix}$$

$O_3 - O_i$ Subtractions (term by term, no premature cancellation)

Define shorthand: $P = 210c_2 + 170c_{23}$ and $Q = 210s_2 + 170s_{23}$.

 $O_3 - O_0$

$$\text{Row 1: } c_1P - 0 = c_1P$$

$$\text{Row 2: } s_1P - 0 = s_1P$$

$$\text{Row 3: } (90 + Q) - 0 = 90 + Q$$

$$O_3 - O_0 = \begin{bmatrix} c_1P \\ s_1P \\ 90 + Q \end{bmatrix}$$

 $O_3 - O_1$

$$\text{Row 1: } c_1P - 0 = c_1P$$

$$\text{Row 2: } s_1P - 0 = s_1P$$

$$\text{Row 3: } (90 + Q) - 90 = Q$$

$$O_3 - O_1 = \begin{bmatrix} c_1P \\ s_1P \\ Q \end{bmatrix}$$

 $O_3 - O_2$

$$\begin{aligned} \text{Row 1: } c_1(210c_2 + 170c_{23}) - 210c_1c_2 &= 210c_1c_2 + 170c_1c_{23} - 210c_1c_2 \\ &= (210c_1c_2 - 210c_1c_2) + 170c_1c_{23} = 170c_1c_{23} \end{aligned}$$

$$\begin{aligned} \text{Row 2: } s_1(210c_2 + 170c_{23}) - 210s_1c_2 &= (210s_1c_2 - 210s_1c_2) + 170s_1c_{23} \\ &= 170s_1c_{23} \end{aligned}$$

$$\begin{aligned} \text{Row 3: } (90 + Q) - (90 + 210s_2) &= (90 - 90) + (210s_2 + 170s_{23} - 210s_2) \\ &= (210s_2 - 210s_2) + 170s_{23} = 170s_{23} \end{aligned}$$

$$O_3 - O_2 = \begin{bmatrix} 170c_1c_{23} \\ 170s_1c_{23} \\ 170s_{23} \end{bmatrix}$$

Step 5: Linear Jacobian Columns $J_{vi} = S(Z_{i-1})(O_3 - O_{i-1})$ **Column 1: $J_{v1} = S(Z_0) \cdot (O_3 - O_0)$**

Full row-by-row multiplication:

$$\text{Row 1: } (0)(c_1P) + (-1)(s_1P) + (0)(90 + Q) = -s_1P$$

$$\text{Row 2: } (1)(c_1P) + (0)(s_1P) + (0)(90 + Q) = c_1P$$

$$\text{Row 3: } (0)(c_1P) + (0)(s_1P) + (0)(90 + Q) = 0$$

$$J_{v1} = \begin{bmatrix} -s_1 P \\ c_1 P \\ 0 \end{bmatrix}$$

Column 2: $J_{v2} = S(Z_1) \cdot (O_3 - O_1)$

$$\text{Row 1: } (0)(c_1 P) + (0)(s_1 P) + (-c_1)(Q) = -c_1 Q$$

$$\text{Row 2: } (0)(c_1 P) + (0)(s_1 P) + (-s_1)(Q) = -s_1 Q$$

$$\text{Row 3: } (c_1)(c_1 P) + (s_1)(s_1 P) + (0)(Q) = c_1^2 P + s_1^2 P = P(c_1^2 + s_1^2) = P$$

$$J_{v2} = \begin{bmatrix} -c_1 Q \\ -s_1 Q \\ P \end{bmatrix}$$

Column 3: $J_{v3} = S(Z_2) \cdot (O_3 - O_2)$

$$\text{Row 1: } (0)(170c_1 c_{23}) + (0)(170s_1 c_{23}) + (-c_1)(170s_{23}) = -170c_1 s_{23}$$

$$\text{Row 2: } (0)(170c_1 c_{23}) + (0)(170s_1 c_{23}) + (-s_1)(170s_{23}) = -170s_1 s_{23}$$

$$\begin{aligned} \text{Row 3: } (c_1)(170c_1 c_{23}) + (s_1)(170s_1 c_{23}) + (0)(170s_{23}) &= 170c_{23}(c_1^2 + s_1^2) \\ &= 170c_{23} \end{aligned}$$

$$J_{v3} = \begin{bmatrix} -170c_1 s_{23} \\ -170s_1 s_{23} \\ 170c_{23} \end{bmatrix}$$

Step 6: Angular Jacobian Columns $J\omega_i = Z_{i-1}$

$$J\omega_1 = Z_0 = [0, 0, 1]^T$$

$$J\omega_2 = Z_1 = [s_1, -c_1, 0]^T$$

$$J\omega_3 = Z_2 = [s_1, -c_1, 0]^T$$

Step 7: Full Geometric Jacobian J (6×3)

Assembling all columns, where $P = 210c_2 + 170c_{23}$ and $Q = 210s_2 + 170s_{23}$:

$$J = \begin{bmatrix} -s_1 P & -c_1 Q & -170c_1 s_{23} & \leftarrow v_x \\ c_1 P & -s_1 Q & -170s_1 s_{23} & \leftarrow v_y \\ 0 & P & 170c_{23} & \leftarrow v_z \\ 0 & s_1 & s_1 & \leftarrow \omega_x \\ 0 & -c_1 & -c_1 & \leftarrow \omega_y \\ 1 & 0 & 0 & \leftarrow \omega_z \end{bmatrix}$$

Analytical (Linear Velocity) Jacobian J_v (3×3)

The upper 3×3 sub-matrix gives the linear velocity mapping $\dot{x} = J_v \dot{\theta}$:

$$J_v = \begin{bmatrix} -s_1P & -c_1Q & -170c_1s_{23} \\ c_1P & -s_1Q & -170s_1s_{23} \\ 0 & P & 170c_{23} \end{bmatrix}$$

Singularity Analysis

A singularity occurs when $\det(J_v) = 0$. The arm cannot generate velocity in all directions at such configurations.

Determinant Calculation — Cofactor Expansion Along Row 3

$$\det(J_v) = 0 \cdot C_{31} + P \cdot C_{32} + 170c_{23} \cdot C_{33}$$

Cofactor C_{32} :

$$\begin{aligned} C_{32} &= (-1)^5 \det[[-s_1P, -170c_1s_{23}], [c_1P, -170s_1s_{23}]] \\ &= -1 \cdot [(-s_1P)(-170s_1s_{23}) - (-170c_1s_{23})(c_1P)] \\ &= -1 \cdot [170s_1^2Ps_{23} + 170c_1^2Ps_{23}] \\ &= -1 \cdot 170Ps_{23}(s_1^2 + c_1^2) = -170Ps_{23} \end{aligned}$$

Cofactor C_{33} :

$$\begin{aligned} C_{33} &= (+1) \det[[-s_1P, -c_1Q], [c_1P, -s_1Q]] \\ &= (-s_1P)(-s_1Q) - (-c_1Q)(c_1P) \\ &= s_1^2PQ + c_1^2PQ = PQ(s_1^2 + c_1^2) = PQ \end{aligned}$$

Substituting:

$$\begin{aligned} \det(J_v) &= P \cdot (-170Ps_{23}) + 170c_{23} \cdot PQ \\ &= 170P(-Ps_{23} + Qc_{23}) \\ &= 170P[c_{23}(210s_2 + 170s_{23}) - s_{23}(210c_2 + 170c_{23})] \\ &= 170P \cdot 210[c_{23}s_2 - s_{23}c_2] \\ &= 170 \cdot 210 \cdot P \cdot \sin(\theta_2 - (\theta_2 + \theta_3)) \\ &= -35700 \cdot P \cdot \sin\theta_3 \end{aligned}$$

$$\det(J_v) = -35700 \cdot (210c_2 + 170c_{23}) \cdot \sin\theta_3$$

Singular Configurations

Singularity 1 — Elbow Singularity:

$$\sin\theta_3 = 0 \Rightarrow \theta_3 = 0^\circ \text{ or } \theta_3 = 180^\circ$$

At $\theta_3=0^\circ$ the arm is fully extended; at $\theta_3=180^\circ$ it is fully folded. Joints 2 and 3 become collinear and lose one independent control direction in the (r,z) plane.

Singularity 2 — Workspace Boundary:

$$P = 0 \Rightarrow 210c_2 + 170c_{23} = 0$$

The end-effector lies on the vertical Z_0 axis ($r = 0$). The azimuthal angle θ_1 is undefined and joint 1 becomes singular. Any θ_1 can produce the same $x=y=0$ position.

Dynamics:

The equation of motion of the 3R manipulator, derived from the Euler-Lagrange formulation, is:

$$M(q)\ddot{q} + B(q)[\dot{q}\dot{q}] + C(q)[\dot{q}^2] + G(q) = \tau$$

where $q = [\theta_1, \theta_2, \theta_3]^T$ is the joint vector, $M(q)$ is the 3×3 symmetric mass matrix, $B(q)$ is the Coriolis matrix, $C(q)$ is the centrifugal matrix, $G(q)$ is the gravity vector, and τ is the joint torque vector.

Inertia Tensor — Link 1

Link 1 is the vertical column. Its dimensions in the Cartesian frame are:

- $a_1 = 55$ mm (along x, horizontal)
- $b_1 = 90$ mm (along y, vertical — this is the link length)
- $c_1 = 60$ mm (along z, out of plane)

The inertia tensor at the centre of mass (COM), derived by direct triple integration over limits $-a_1/2$ to $a_1/2$ in x, $-b_1/2$ to $b_1/2$ in y, $-c_1/2$ to $c_1/2$ in z:

General derivation of I_{xx} (same procedure for other components):

$$\begin{aligned} I_{xx} &= \rho \int \int \int (y^2 + z^2) dV \\ &= \rho \int_{-a/2}^{a/2} \int_{-b/2}^{b/2} \int_{-c/2}^{c/2} (y^2 + z^2) dz dy dx \\ &= \rho \cdot a \cdot \int_{-b/2}^{b/2} [y^2 z + z^3/3]_{-c/2}^{c/2} dy \\ &= \rho \cdot a \cdot \int_{-b/2}^{b/2} (y^2 c + c^3/12) dy \\ &= \rho \cdot a \cdot [cy^3/3 + c^3 y/12]_{-b/2}^{b/2} \\ &= \rho \cdot a \cdot (cb^3/12 + c^3 b/12) = \rho \cdot abc \cdot (b^2 + c^2)/12 = m(b^2 + c^2)/12 \end{aligned}$$

Applying to Link 1 ($a_1=55$, $b_1=90$, $c_1=60$, mass m_1):

$$\begin{aligned} I_{1xx_com} &= m_1(b_1^2 + c_1^2)/12 = m_1(90^2 + 60^2)/12 = m_1(8100 + 3600)/12 \\ &= 11700m_1/12 = 975m_1 \text{ mm}^2 \end{aligned}$$

$$I_{1yy_com} = m_1(a_1^2 + c_1^2)/12 = m_1(55^2 + 60^2)/12 = m_1(3025 + 3600)/12 \\ = 6625m_1/12 = 552.08m_1 \text{ mm}^2$$

$$I_{1zz_com} = m_1(a_1^2 + b_1^2)/12 = m_1(55^2 + 90^2)/12 = m_1(3025 + 8100)/12 \\ = 11125m_1/12 = 927.08m_1 \text{ mm}^2$$

$$I_{1_com} = \begin{bmatrix} 975m_1 & 0 & 0 \\ 0 & 552.08m_1 & 0 \\ 0 & 0 & 927.08m_1 \end{bmatrix}$$

Parallel-Axis Theorem — shift from COM to bottom face (motor mount). The COM shifts downward along y by $b_1/2 = 45 \text{ mm}$, so $d = [0, 45, 0]^T$:

$$I_{1xx} = I_{1xx_com} + m_1 \cdot (45^2) = 975m_1 + 2025m_1 = 3000m_1 \text{ mm}^2$$

$$I_{1yy} = I_{1yy_com} + 0 = 552.08m_1 \text{ mm}^2 \quad (\text{shift is along } y \\ \Rightarrow I_{yy} \text{ unchanged})$$

$$I_{1zz} = I_{1zz_com} + m_1 \cdot (45^2) = 927.08m_1 + 2025m_1 = 2952.08m_1 \text{ mm}^2$$

$$I_1 = \begin{bmatrix} 3000m_1 & 0 & 0 \\ 0 & 552.08m_1 & 0 \\ 0 & 0 & 2952.08m_1 \end{bmatrix}$$

Scalar effective inertia for dynamics: Joint 1 rotates about the y-axis, so the rotation axis normal is $n_1 = [0, 1, 0]^T$:

$$I_1 = n_1^T \cdot I_1 \cdot n_1 = I_{1yy_com} = 552.08 m_1 \text{ mm}^2$$

Inertia Tensor — Link 2

Link 2 is horizontal. Dimensions: $a_2=210 \text{ mm}$ (x, length), $b_2=55 \text{ mm}$ (y), $c_2=60 \text{ mm}$ (z).

$$I_{2xx_com} = m_2(b_2^2 + c_2^2)/12 = m_2(55^2 + 60^2)/12 = m_2(3025 + 3600)/12 \\ = 552.08m_2 \text{ mm}^2$$

$$I_{2yy_com} = m_2(a_2^2 + c_2^2)/12 = m_2(210^2 + 60^2)/12 = m_2(44100 + 3600)/12 \\ = 47700m_2/12 = 3975m_2 \text{ mm}^2$$

$$I_{2zz_com} = m_2(a_2^2 + b_2^2)/12 = m_2(210^2 + 55^2)/12 = m_2(44100 + 3025)/12 \\ = 47125m_2/12 = 3927.08m_2 \text{ mm}^2$$

PAT — shift from COM to left end (Joint 2 mount) along x by $a_2/2 = 105 \text{ mm}$. $d = [105, 0, 0]^T$:

$$I_{2xx} = I_{2xx_com} = 552.08m_2 \text{ mm}^2 \quad (\text{shift along } x \Rightarrow I_{xx} \text{ unchanged})$$

$$I_{2yy} = I_{2yy_com} + m_2 \cdot (105^2) = 3975m_2 + 11025m_2 = 15000m_2 \text{ mm}^2$$

$$I_{2zz} = I_{2zz_com} + m_2 \cdot (105^2) = 3927.08m_2 + 11025m_2 = 14952.08m_2 \text{ mm}^2$$

$$I_2 = \begin{bmatrix} 552.08m_2 & 0 & 0 \\ 0 & 15000m_2 & 0 \\ 0 & 0 & 14952.08m_2 \end{bmatrix}$$

Scalar effective inertia: Joint 2 rotates about the z-axis, $n_2 = [0,0,1]^T$:

$$I_2 = n_2^T \cdot I_2 \cdot n_2 = I_{2zz} = 14952.08 m_2 \text{ mm}^2$$

Inertia Tensor — Link 3

Link 3 is horizontal. Dimensions: $a_3=170$ mm (x), $b_3=55$ mm (y), $c_3=60$ mm (z).

$$I_{3xx_com} = m_3(55^2 + 60^2)/12 = 552.08m_3 \text{ mm}^2$$

$$I_{3yy_com} = m_3(170^2 + 60^2)/12 = m_3(28900 + 3600)/12 = 32500m_3/12 = 2708.33m_3 \text{ mm}^2$$

$$I_{3zz_com} = m_3(170^2 + 55^2)/12 = m_3(28900 + 3025)/12 = 31925m_3/12 = 2660.42m_3 \text{ mm}^2$$

PAT — shift from COM to left end (Joint 3 mount) along x by $a_3/2 = 85$ mm. $d = [85,0,0]^T$:

$$I_{3xx} = 552.08m_3 \text{ mm}^2 \quad (\text{shift along } x \Rightarrow I_{xx} \text{ unchanged})$$

$$I_{3yy} = 2708.33m_3 + m_3 \cdot (85^2) = 2708.33m_3 + 7225m_3 = 9933.33m_3 \text{ mm}^2$$

$$I_{3zz} = 2660.42m_3 + m_3 \cdot (85^2) = 2660.42m_3 + 7225m_3 = 9885.42m_3 \text{ mm}^2$$

$$I_3 = \begin{bmatrix} 552.08m_3 & 0 & 0 \\ 0 & 9933.33m_3 & 0 \\ 0 & 0 & 9885.42m_3 \end{bmatrix}$$

Scalar effective inertia: Joint 3 also rotates about z-axis, $n_3 = [0,0,1]^T$:

$$I_3 = I_{3zz} = 9885.42 m_3 \text{ mm}^2$$

Table 4: Summary of Inertia Tensor Values at Motor Mount Points

Link	I _{xx} (mm ²)	I _{yy} (mm ²)	I _{zz} (mm ²)	Scalar I _i used in dynamics
Link 1	3000 m ₁	552.08 m ₁ (=I ₁)	2952.08 m ₁	I ₁ = 552.08 m ₁ (rot. about y)
Link 2	552.08 m ₂	15000 m ₂	14952.08 m ₂ (=I ₂)	I ₂ = 14952.08 m ₂ (rot. about z)
Link 3	552.08 m ₃	9933.33 m ₃	9885.42 m ₃ (=I ₃)	I ₃ = 9885.42 m ₃ (rot. about z)

Centre of Mass Positions

COM half-lengths: $lc_1 = b_1/2 = 45 \text{ mm}$, $lc_2 = a_2/2 = 105 \text{ mm}$, $lc_3 = a_3/2 = 85 \text{ mm}$.

COM positions in the DH base frame:

$$C_1 = \begin{bmatrix} 0 \\ 0 \\ lc_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 45 \end{bmatrix}$$

(C_1 lies on the Z_0 axis — joint 1 rotation produces no linear velocity at this COM)

$$C_2 = \begin{bmatrix} lc_2 c_1 c_2 \\ lc_2 s_1 c_2 \\ d_1 + lc_2 s_2 \end{bmatrix} = \begin{bmatrix} 105 c_1 c_2 \\ 105 s_1 c_2 \\ 90 + 105 s_2 \end{bmatrix}$$

Shorthand: $Pc = l_2 c_2 + l_3 c_{23} = 210 c_2 + 85 c_{23}$, $Qc = l_2 s_2 + l_3 s_{23} = 210 s_2 + 85 s_{23}$

$$C_3 = \begin{bmatrix} c_1 \cdot Pc \\ s_1 \cdot Pc \\ 90 + Qc \end{bmatrix} = \begin{bmatrix} c_1(210 c_2 + 85 c_{23}) \\ s_1(210 c_2 + 85 c_{23}) \\ 90 + 210 s_2 + 85 s_{23} \end{bmatrix}$$

COM Jacobians

The COM Jacobian for link i uses the same $S(Z)$ method but targets C_i instead of O_3 . Columns after joint i are zero (joints beyond i do not move COM of link i).

JvC_1 and $J\omega C_1$ (Link 1 — column 1 only, rest zero)

Column 1: $Jv_1 C_1 = S(Z_0)(C_1 - O_0)$

$$C_1 - O_0 = [0, 0, 45]^T - [0, 0, 0]^T = [0, 0, 45]^T$$

$$\text{Row 1: } (0)(0) + (-1)(0) + (0)(45) = 0$$

$$\text{Row 2: } (1)(0) + (0)(0) + (0)(45) = 0$$

$$\text{Row 3: } (0)(0) + (0)(0) + (0)(45) = 0$$

C_1 lies on the Z_0 axis $\Rightarrow$ joint 1 rotation produces zero linear velocity at C_1 .

$$JvC_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad J\omega C_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

JvC₂ and JωC₂ (Link 2 — columns 1 and 2; column 3 = 0)

Column 1: $Jv_{1_}C_2 = S(Z_0)(C_2 - O_0)$

$$C_2 - O_0 = [105c_1c_2, 105s_1c_2, 90 + 105s_2]^T$$

$$\text{Row 1: } (0)(105c_1c_2) + (-1)(105s_1c_2) + (0)(90 + 105s_2) = -105s_1c_2$$

$$\text{Row 2: } (1)(105c_1c_2) + (0)(105s_1c_2) + (0)(90 + 105s_2) = 105c_1c_2$$

$$\text{Row 3: } (0)(105c_1c_2) + (0)(105s_1c_2) + (0)(90 + 105s_2) = 0$$

Column 2: $Jv_{2_}C_2 = S(Z_1)(C_2 - O_1)$

$$C_2 - O_1 = [105c_1c_2 - 0, 105s_1c_2 - 0, (90 + 105s_2) - 90]^T \\ = [105c_1c_2, 105s_1c_2, 105s_2]^T$$

$$\text{Row 1: } (0)(105c_1c_2) + (0)(105s_1c_2) + (-c_1)(105s_2) = -105c_1s_2$$

$$\text{Row 2: } (0)(105c_1c_2) + (0)(105s_1c_2) + (-s_1)(105s_2) = -105s_1s_2$$

$$\text{Row 3: } (c_1)(105c_1c_2) + (s_1)(105s_1c_2) + (0)(105s_2) = 105c_2(c_1^2 + s_1^2) = 105c_2$$

$$JvC_2 = \begin{bmatrix} -105s_1c_2 & -105c_1s_2 & 0 \\ 105c_1c_2 & -105s_1s_2 & 0 \\ 0 & 105c_2 & 0 \end{bmatrix}$$

$$J\omega C_2 = \begin{bmatrix} 0 & s_1 & 0 \\ 0 & -c_1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

JvC₃ and JωC₃ (Link 3 — all three columns)

Column 1: $Jv_{1_}C_3 = S(Z_0)(C_3 - O_0)$

$$C_3 - O_0 = [c_1Pc, s_1Pc, 90 + Qc]^T$$

$$\text{Row 1: } (-1)(s_1Pc) = -s_1Pc$$

$$\text{Row 2: } (1)(c_1Pc) = c_1Pc$$

$$\text{Row 3: } 0$$

Column 2: $Jv_{2_}C_3 = S(Z_1)(C_3 - O_1)$

$$C_3 - O_1 = [c_1Pc, s_1Pc, Qc]^T$$

$$\text{Row 1: } (0)(c_1Pc) + (0)(s_1Pc) + (-c_1)(Qc) = -c_1Qc$$

$$\text{Row 2: } (0)(c_1Pc) + (0)(s_1Pc) + (-s_1)(Qc) = -s_1Qc$$

$$\text{Row 3: } (c_1)(c_1Pc) + (s_1)(s_1Pc) + (0)(Qc) = Pc(c_1^2 + s_1^2) = Pc$$

Column 3: $Jv_{3_}C_3 = S(Z_2)(C_3 - O_2)$

$$C_3 - O_2 = [85c_1c_{23}, 85s_1c_{23}, 85s_{23}]^T$$

$$\begin{aligned}
\text{Row 1: } & (0)(85c_1c_{23}) + (0)(85s_1c_{23}) + (-c_1)(85s_{23}) = -85c_1s_{23} \\
\text{Row 2: } & (0)(85c_1c_{23}) + (0)(85s_1c_{23}) + (-s_1)(85s_{23}) = -85s_1s_{23} \\
\text{Row 3: } & (c_1)(85c_1c_{23}) + (s_1)(85s_1c_{23}) + (0)(85s_{23}) = 85c_{23}(c_1^2 + s_1^2) = 85c_{23}
\end{aligned}$$

$$JvC_3 = \begin{bmatrix} -s_1Pc & -c_1Qc & -85c_1s_{23} \\ c_1Pc & -s_1Qc & -85s_1s_{23} \\ 0 & Pc & 85c_{23} \end{bmatrix}$$

$$J\omega C_3 = \begin{bmatrix} 0 & s_1 & s_1 \\ 0 & -c_1 & -c_1 \\ 1 & 0 & 0 \end{bmatrix}$$

Mass Matrix M(q)

The mass matrix is assembled from contributions of all three links:

$$M(q) = \sum_i [m_i JvC_i^T JvC_i + I_i J\omega C_i^T J\omega C_i]$$

Link 1 Contribution

Since $JvC_1 = 0$, the translational part vanishes. Rotational part:

$$\begin{aligned}
J\omega C_1^T J\omega C_1 &= [1,0,0; 0,0,0; 0,0,0] \\
M_{\text{link1}} &= I_1 \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}
\end{aligned}$$

Link 2 Contribution

Translational part $m_2 JvC_2^T JvC_2$:

$$\begin{aligned}
(1,1): & (-105s_1c_2)^2 + (105c_1c_2)^2 + 0 = 105^2c_2^2(s_1^2 + c_1^2) = 105^2c_2^2 = lc_2^2c_2^2 \\
(2,2): & (-105c_1s_2)^2 + (-105s_1s_2)^2 + (105c_2)^2 = 105^2s_2^2 + 105^2c_2^2 = 105^2 \\
& = lc_2^2 \\
(3,3): & 0 \\
(1,2): & (-105s_1c_2)(-105c_1s_2) + (105c_1c_2)(-105s_1s_2) \\
& = 105^2s_1c_1c_2s_2 - 105^2c_1s_1s_2c_2 = 0
\end{aligned}$$

Rotational part $I_2 J\omega C_2^T J\omega C_2$:

$$\begin{aligned}
J\omega C_2^T J\omega C_2 \text{ (1,1): } & 0^2 + 0^2 + 1^2 = 1, \quad (2,2): s_1^2 + c_1^2 = 1, \quad (3,3): 0, \\
& \text{off-diag: } 0
\end{aligned}$$

$$M_{link2} = m_2 \begin{bmatrix} lc_2^2 c_2^2 & 0 & 0 \\ 0 & lc_2^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} + I_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Link 3 Contribution

Translational part $m_3 JvC_3^T JvC_3$:

$$\begin{aligned} (1,1): & (-s_1 Pc)^2 + (c_1 Pc)^2 + 0 = Pc^2 \\ (2,2): & (-c_1 Qc)^2 + (-s_1 Qc)^2 + Pc^2 = Qc^2 + Pc^2 \text{ [note: } Pc^2 + Qc^2 \\ & = 210^2 + 2 \cdot 210 \cdot 85c_3 + 85^2] \\ (3,3): & (-85c_1 s_{23})^2 + (-85s_1 s_{23})^2 + (85c_{23})^2 = 85^2 s_{23}^2 + 85^2 c_{23}^2 = 85^2 = lc_3^2 \\ (2,3): & (-c_1 Qc)(-85c_1 s_{23}) + (-s_1 Qc)(-85s_1 s_{23}) + (Pc)(85c_{23}) \\ & = 85Qc s_{23}(c_1^2 + s_1^2) + 85Pcc_{23} \\ & = 85[Qc s_{23} + Pcc_{23}] \\ & = 85[210(s_2 s_{23} + c_2 c_{23}) + 85(s_{23}^2 + c_{23}^2)] \\ & = 85[210 \cos \theta_3 + 85] = 17850c_3 + 7225 = lc_3(l_2 c_3 + lc_3) \\ (1,2) & = 0, \quad (1,3) = 0 \end{aligned}$$

Rotational part $I_3 J\omega C_3^T J\omega C_3$:

$$\begin{aligned} (1,1): & 1, \quad (2,2): s_1^2 + c_1^2 = 1, \quad (3,3): s_1^2 + c_1^2 = 1, \\ (2,3) & = s_1 s_1 + c_1 c_1 = 1 \end{aligned}$$

$$M_{link3} = m_3 \begin{bmatrix} Pc^2 & 0 & 0 \\ 0 & Pc^2 + Qc^2 & 17850c_3 + 7225 \\ 0 & 17850c_3 + 7225 & 85^2 \end{bmatrix} + I_3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Final Mass Matrix $M(q) = M_{link1} + M_{link2} + M_{link3}$

$$M(q) = \begin{bmatrix} M_{11} & 0 & 0 \\ 0 & M_{22} & M_{23} \\ 0 & M_{23} & M_{33} \end{bmatrix}$$

where:

$$\begin{aligned} M_{11} & = I_1 + m_2 lc_2^2 c_2^2 + I_2 + m_3(l_2 c_2 + lc_3 c_{23})^2 + I_3 \\ M_{22} & = m_2 lc_2^2 + I_2 + m_3(l_2^2 + lc_3^2 + 2l_2 lc_3 c_3) + I_3 \\ M_{33} & = m_3 lc_3^2 + I_3 \\ M_{23} & = M_{32} = m_3(lc_3^2 + l_2 lc_3 c_3) + I_3 \\ M_{12} & = M_{13} = 0 \end{aligned}$$

Kinetic Energy K

$$K = \frac{1}{2} \theta^T M(q) \theta$$

Expanding step by step. First compute $M(q)\theta$:

$$M(q)\theta = [M_{11}\theta_1, M_{22}\theta_2 + M_{23}\theta_3, M_{23}\theta_2 + M_{33}\theta_3]^T$$

Then multiply by θ^T :

$$\begin{aligned} K &= \frac{1}{2}[\theta_1(M_{11}\theta_1) + \theta_2(M_{22}\theta_2 + M_{23}\theta_3) + \theta_3(M_{23}\theta_2 + M_{33}\theta_3)] \\ &= \frac{1}{2}M_{11}\theta_1^2 + \frac{1}{2}M_{22}\theta_2^2 + \frac{1}{2}M_{23}\theta_2\theta_3 + \frac{1}{2}M_{23}\theta_3\theta_2 + \frac{1}{2}M_{33}\theta_3^2 \\ &= \frac{1}{2}M_{23}\theta_2\theta_3 + \frac{1}{2}M_{23}\theta_2\theta_3 = M_{23}\theta_2\theta_3 \text{ [combining identical cross terms]} \end{aligned}$$

$$K = \frac{1}{2}M_{11}\theta_1^2 + \frac{1}{2}M_{22}\theta_2^2 + M_{23}\theta_2\theta_3 + \frac{1}{2}M_{33}\theta_3^2$$

Potential Energy U

Taking heights as the y (vertical) coordinate of each COM:

$$h_1 = lc_1 = 45 \text{ mm (constant)}$$

$$h_2 = d_1 + lc_2 s_2 = 90 + 105s_2$$

$$h_3 = d_1 + l_2 s_2 + lc_3 s_{23} = 90 + 210s_2 + 85s_{23}$$

$$U = m_1 g lc_1 + m_2 g (d_1 + lc_2 s_2) + m_3 g (d_1 + l_2 s_2 + lc_3 s_{23})$$

Lagrangian

$$L = K - U$$

$$\begin{aligned} L &= \frac{1}{2}M_{11}\theta_1^2 + \frac{1}{2}M_{22}\theta_2^2 + M_{23}\theta_2\theta_3 + \frac{1}{2}M_{33}\theta_3^2 \\ &\quad - m_1 g lc_1 - m_2 g (d_1 + lc_2 s_2) - m_3 g (d_1 + l_2 s_2 + lc_3 s_{23}) \end{aligned}$$

Derivation of τ_1

Lagrange equation:

$$\tau_1 = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) - \frac{\partial L}{\partial \theta_1}$$

Step 1: $\partial L / \partial \theta_1$

Neither K nor U contain θ_1 explicitly (M_{11} , M_{22} , M_{23} , M_{33} depend only on θ_2 , θ_3 ; potential energy has no θ_1). Therefore:

$$\frac{\partial L}{\partial \theta_1} = 0$$

Step 2: $\partial L / \partial \theta_i$

$$\partial L / \partial \theta_1 = M_{11} \dot{\theta}_1$$

Step 3: $d/dt(M_{11}\dot{\theta}_1)$ — Product Rule

$$\frac{d}{dt}(M_{11}\dot{\theta}_1) = \dot{M}_{11}\dot{\theta}_1 + M_{11}\ddot{\theta}_1$$

Step 4: Compute $\dot{M}_{11}$

Recall $M_{11} = I_1 + I_2 + I_3 + m_2 l c_2^2 c_2^2 + m_3 (l_2 c_2 + l c_3 c_{23})^2$. Constants vanish. Expanding $(l_2 c_2 + l c_3 c_{23})^2$:

$$= l_2^2 c_2^2 + 2 l_2 l c_3 c_2 c_{23} + l c_3^2 c_{23}^2$$

Variable terms: $m_2 l c_2^2 c_2^2$, $m_3 l_2^2 c_2^2$, $2 m_3 l_2 l c_3 c_2 c_{23}$, $m_3 l c_3^2 c_{23}^2$

Differentiating each term with $\dot{c}_2 = -s_2 \dot{\theta}_2$ and $\dot{c}_{23} = -s_{23}(\dot{\theta}_2 + \dot{\theta}_3)$:

$$d/dt(m_2 l c_2^2 c_2^2) = -2 m_2 l c_2^2 c_2 s_2 \dot{\theta}_2$$

$$d/dt(m_3 l_2^2 c_2^2) = -2 m_3 l_2^2 c_2 s_2 \dot{\theta}_2$$

$$d/dt(2 m_3 l_2 l c_3 c_2 c_{23}) = 2 m_3 l_2 l c_3 [\dot{c}_2 c_{23} + c_2 \dot{c}_{23}]$$

$$= 2 m_3 l_2 l c_3 [(-s_2 \dot{\theta}_2) c_{23} + c_2 (-s_{23})(\dot{\theta}_2 + \dot{\theta}_3)]$$

$$= 2 m_3 l_2 l c_3 [-s_2 c_{23} \dot{\theta}_2 - c_2 s_{23} \dot{\theta}_2 - c_2 s_{23} \dot{\theta}_3]$$

$$= -2 m_3 l_2 l c_3 [(s_2 c_{23} + c_2 s_{23}) \dot{\theta}_2 + c_2 s_{23} \dot{\theta}_3] \text{ using identity } s_2 c_{23} + c_2 s_{23} = s_{23}$$

$$= -2 m_3 l_2 l c_3 [s_{23} \dot{\theta}_2 + c_2 s_{23} \dot{\theta}_3]$$

$$d/dt(m_3 l c_3^2 c_{23}^2) = m_3 l c_3^2 \cdot 2 c_{23} \cdot (-s_{23})(\dot{\theta}_2 + \dot{\theta}_3) = -2 m_3 l c_3^2 c_{23} s_{23} (\dot{\theta}_2 + \dot{\theta}_3)$$

Collecting all terms:

$$\dot{M}_{11} = [-2 m_2 l c_2^2 c_2 s_2 - 2 m_3 l_2^2 c_2 s_2 - 2 m_3 l_2 l c_3 s_{23} - 2 m_3 l c_3^2 c_{23} s_{23}] \dot{\theta}_2$$

$$+ [-2 m_3 l_2 l c_3 c_2 s_{23} - 2 m_3 l c_3^2 c_{23} s_{23}] \dot{\theta}_3$$

$$= -2 m_3 (l_2 c_2 + l c_3 c_{23}) [l_2 s_2 \dot{\theta}_2 + l c_3 s_{23} (\dot{\theta}_2 + \dot{\theta}_3)] - 2 m_2 l c_2^2 c_2 s_2 \dot{\theta}_2$$

Step 5: Final τ_1

$$\tau_1 = M_{11} \ddot{\theta}_1$$

$$\begin{aligned}
& -2m_2lc_2^2c_2s_2 \cdot \theta_1\theta_2 \\
& -2m_3(l_2c_2 + lc_3c_{23})l_2s_2 \cdot \theta_1\theta_2 \\
& -2m_3(l_2c_2 + lc_3c_{23})lc_3s_{23} \cdot \theta_1\theta_2 \\
& -2m_3(l_2c_2 + lc_3c_{23})lc_3s_{23} \cdot \theta_1\theta_3
\end{aligned}$$

Derivation of τ_2

Step 1: $\partial L/\partial \theta_2$

$$\partial L/\partial \theta_2 = M_{22}\theta_2 + M_{23}\theta_3$$

Step 2: $d/dt(M_{22}\theta_2 + M_{23}\theta_3)$ — Product Rule on each term

Differentiating $M_{22} = m_2lc_2^2 + I_2 + m_3(l_2^2 + lc_3^2 + 2l_2lc_3c_3) + I_3$: only $2l_2lc_3c_3$ varies.

$$\begin{aligned}
\dot{M}_{22} &= m_3 \cdot 2l_2lc_3 \cdot (-s_3\theta_3) = -2m_3l_2lc_3s_3\theta_3 \\
d/dt(M_{22}\theta_2) &= \dot{M}_{22}\theta_2 + M_{22}\dot{\theta}_2 = -2m_3l_2lc_3s_3\theta_2\theta_3 + M_{22}\ddot{\theta}_2
\end{aligned}$$

Differentiating $M_{23} = m_3(lc_3^2 + l_2lc_3c_3) + I_3$: only $l_2lc_3c_3$ varies.

$$\begin{aligned}
\dot{M}_{23} &= m_3l_2lc_3 \cdot (-s_3\theta_3) = -m_3l_2lc_3s_3\theta_3 \\
d/dt(M_{23}\theta_3) &= \dot{M}_{23}\theta_3 + M_{23}\dot{\theta}_3 = -m_3l_2lc_3s_3\theta_3^2 + M_{23}\ddot{\theta}_3
\end{aligned}$$

Combining:

$$d/dt(\partial L/\partial \theta_2) = M_{22}\ddot{\theta}_2 + M_{23}\ddot{\theta}_3 - 2m_3l_2lc_3s_3\theta_2\theta_3 - m_3l_2lc_3s_3\theta_3^2$$

Step 3: $\partial L/\partial \theta_2$

Only M_{11} depends on θ_2 :

$$\begin{aligned}
\partial K/\partial \theta_2 &= \frac{1}{2}(\partial M_{11}/\partial \theta_2)\theta_1^2 \\
\partial M_{11}/\partial \theta_2 &= -2m_2lc_2^2c_2s_2 - 2m_3(l_2c_2 + lc_3c_{23})(l_2s_2 + lc_3s_{23}) \\
\partial U/\partial \theta_2 &= m_2g lc_2c_2 + m_3g(l_2c_2 + lc_3c_{23})
\end{aligned}$$

Step 4: Final τ_2

$$\begin{aligned}
\tau_2 &= M_{22}\ddot{\theta}_2 + M_{23}\ddot{\theta}_3 \\
&\quad - 2m_3l_2lc_3s_3 \cdot \theta_2\theta_3 \\
&\quad - m_3l_2lc_3s_3 \cdot \theta_3^2 \\
&\quad + [m_2lc_2^2c_2s_2 + m_3(l_2c_2 + lc_3c_{23})(l_2s_2 + lc_3s_{23})] \cdot \theta_1^2 \\
&\quad + m_2g lc_2c_2 + m_3g(l_2c_2 + lc_3c_{23})
\end{aligned}$$

Derivation of τ_3

Step 1: $\partial L / \partial \dot{\theta}_3$

$$\partial L / \partial \dot{\theta}_3 = M_{23}\dot{\theta}_2 + M_{33}\dot{\theta}_3$$

Step 2: $d/dt(M_{23}\dot{\theta}_2 + M_{33}\dot{\theta}_3)$

$$\begin{aligned} d/dt(M_{23}\dot{\theta}_2) &= \dot{M}_{23}\dot{\theta}_2 + M_{23}\ddot{\theta}_2 = -m_3l_2lc_3s_3\dot{\theta}_2\dot{\theta}_3 + M_{23}\ddot{\theta}_2 \\ d/dt(M_{33}\dot{\theta}_3) &= \dot{M}_{33}\dot{\theta}_3 + M_{33}\ddot{\theta}_3 \quad [M_{33} = m_3lc_3^2 + I_3 = \text{constant} \Rightarrow \dot{M}_{33} = 0] = M_{33}\ddot{\theta}_3 \\ d/dt(\partial L / \partial \dot{\theta}_3) &= M_{23}\ddot{\theta}_2 + M_{33}\ddot{\theta}_3 - m_3l_2lc_3s_3\dot{\theta}_2\dot{\theta}_3 \end{aligned}$$

Step 3: $\partial L / \partial \theta_3$

Three elements M_{11} , M_{22} , M_{23} depend on θ_3 :

$$\begin{aligned} \partial K / \partial \theta_3 &= \frac{1}{2}(\partial M_{11} / \partial \theta_3)\dot{\theta}_1^2 + \frac{1}{2}(\partial M_{22} / \partial \theta_3)\dot{\theta}_2^2 + (\partial M_{23} / \partial \theta_3)\dot{\theta}_2\dot{\theta}_3 \\ \partial M_{11} / \partial \theta_3 &= -2m_3lc_3(l_2c_2 + lc_3c_{23})s_{23} \\ \partial M_{22} / \partial \theta_3 &= -2m_3l_2lc_3s_3 \\ \partial M_{23} / \partial \theta_3 &= -m_3l_2lc_3s_3 \\ \partial U / \partial \theta_3 &= m_3glc_3c_{23} \\ \partial L / \partial \theta_3 &= -m_3lc_3(l_2c_2 + lc_3c_{23})s_{23}\dot{\theta}_1^2 - m_3l_2lc_3s_3\dot{\theta}_2^2 - m_3l_2lc_3s_3\dot{\theta}_2\dot{\theta}_3 \\ &\quad - m_3glc_3c_{23} \end{aligned}$$

Step 4: Combine — note $\dot{\theta}_2\dot{\theta}_3$ terms cancel

From Step 2: $-m_3l_2lc_3s_3\dot{\theta}_2\dot{\theta}_3$

From Step 3 (negated): $+m_3l_2lc_3s_3\dot{\theta}_2\dot{\theta}_3$

Sum = 0 $\Rightarrow$ $\dot{\theta}_2\dot{\theta}_3$ terms cancel exactly. $\checkmark$

Step 5: Final τ_3

$$\begin{aligned} \tau_3 &= M_{23}\ddot{\theta}_2 + M_{33}\ddot{\theta}_3 \\ &\quad + m_3lc_3(l_2c_2 + lc_3c_{23})s_{23} \cdot \dot{\theta}_1^2 \\ &\quad + m_3l_2lc_3s_3 \cdot \dot{\theta}_2^2 \\ &\quad + m_3glc_3c_{23} \end{aligned}$$

Extraction of $\mathbf{M}(\mathbf{q})$, $\mathbf{B}(\mathbf{q})$, $\mathbf{C}(\mathbf{q})$, $\mathbf{G}(\mathbf{q})$

Velocity vectors: $[\dot{\mathbf{q}}] = [\dot{\theta}_1\dot{\theta}_2, \dot{\theta}_1\dot{\theta}_3, \dot{\theta}_2\dot{\theta}_3]^T$ (mixed), $[\mathbf{q}^2] = [\theta_1^2, \theta_2^2, \theta_3^2]^T$ (squared)

Mass Matrix $M(q)$ — from $\theta\ddot{a}$ coefficients:

$$M(q) = \begin{bmatrix} M_{11} & 0 & 0 \\ 0 & M_{22} & M_{23} \\ 0 & M_{23} & M_{33} \end{bmatrix}$$

Coriolis Matrix $B(q)$ — from mixed velocity products:

$$B(q) = \begin{bmatrix} -2(m_2lc_2^2c_2s_2 + m_3Pc \cdot Qs) & -2m_3Pc \cdot lc_3s_{23} & 0 \\ 0 & 0 & -2m_3l_2lc_3s_3 \\ 0 & 0 & 0 \end{bmatrix}$$

where $Pc = l_2c_2 + lc_3c_{23}$ and $Qs = l_2s_2 + lc_3s_{23}$

Centrifugal Matrix $C(q)$ — from squared velocity terms:

$$C(q) = \begin{bmatrix} 0 & 0 & 0 \\ (m_2lc_2^2c_2s_2 + m_3Pc \cdot Qs) & 0 & -m_3l_2lc_3s_3 \\ m_3lc_3Pc \cdot s_{23} & m_3l_2lc_3s_3 & 0 \end{bmatrix}$$

Gravity Vector $G(q)$:

$$G(q) = \begin{bmatrix} 0 \\ m_2glc_2c_2 + m_3g(l_2c_2 + lc_3c_{23}) \\ m_3glc_3c_{23} \end{bmatrix}$$

Note: $G_1 = 0$ because joint 1 rotates about the vertical axis and performs no work against gravity regardless of θ_1 . This is physically consistent and serves as a verification checkpoint. **Control**

Basic Revision

PID Controller

A Proportional-Integral-Derivative (PID) controller produces a control output $u(t)$ from joint angle error $e(t) = \theta_d(t) - \theta(t)$:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de}{dt}$$

In discrete form (implemented on ESP32 at sampling period T_s):

$$u[k] = K_p e[k] + K_i T_s \sum e[k] + K_d \frac{e[k] - e[k-1]}{T_s}$$

State-Space Representation

Single-joint dynamics in state-space form with state $x = [\theta, \dot{\theta}]^T$:

$$\dot{x} = Ax + Bu, \quad y = Cx$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -b/J \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1/J \end{bmatrix}, \quad C = [1 \quad 0]$$

Robot Control Strategies

Joint-Space Control (Independent PID Loops)

The simplest strategy treats each joint independently. The ESP32 maps target angle to PWM pulse width:

$$\text{Pulse width } (\mu s) = 500 + \frac{\theta_i}{180^\circ} \times 2000$$

Ignores coupling between joints. Suitable for slow, lightly loaded motion.

Computed Torque Method (Model-Based)

The full dynamic model is inverted to cancel nonlinearities:

$$\tau = M(q)(\ddot{\theta}_d + K_d\dot{e} + K_p e) + B(q)[\dot{q}] + C(q)[q^2] + G(q)$$

This reduces the closed-loop system to three decoupled linear second-order systems. Requires accurate mass values.

Task-Space Control

Position error in Cartesian space drives the joints via the Jacobian inverse:

$$\dot{\theta} = J^v{}^+ (\dot{x}_d + K_p(x_d - x_e))$$

The pseudo-inverse $J_v^+ = J_v^T(J_v J_v^T)^{-1}$ is used near singularities. Preferred for Cartesian straight-line trajectories.